CP2.5 — Polar Coordinates

1. Introduction to Polar Coordinates

A point P has **polar coordinates** (r, θ) where:

- r = distance from origin (pole)
- θ = angle measured anticlockwise from initial line (positive x-axis)

Convention: $r \geq 0$, $-\pi < \theta \leq \pi$ (or $0 \leq \theta < 2\pi$).

1.1 Converting Between Polar and Cartesian

$$x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}$$

Example: Convert $(3, \pi/3)$ to Cartesian.
$x = 3\cos(\pi/3) = 3/2$, $y = 3\sin(\pi/3) = 3\sqrt{3}/2$.

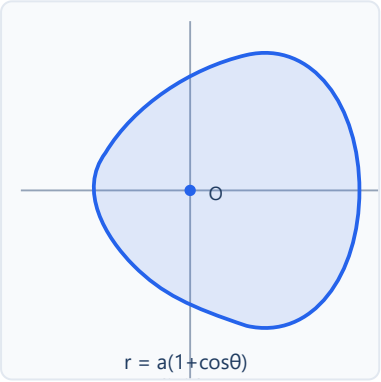
Example: Convert $(-1, \sqrt{3})$ to polar.
$r = \sqrt{1+3} = 2$, $\theta = \arctan(\sqrt{3}/(-1))$. Point is in Q2, so $\theta = \pi - \pi/3 = 2\pi/3$.

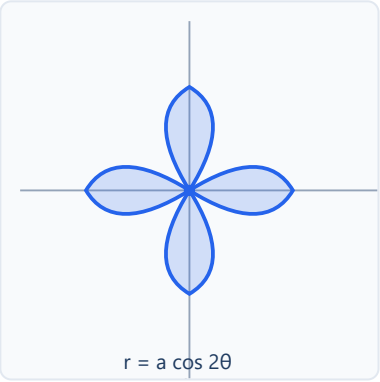
2. Sketching Polar Curves

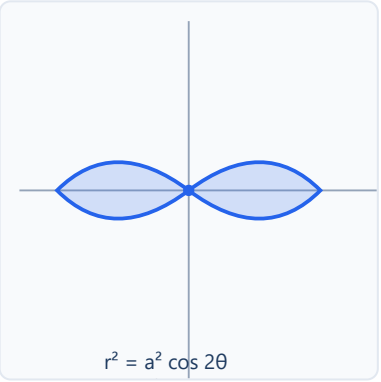
2.1 Key Polar Curves — Summary Table

Equation	Curve	Key features

$r = a$	Circle	Centre at origin, radius a
$r = 2a\cos\theta$	Circle	Centre $(a, 0)$, radius a; passes through origin
$r = 2a\sin\theta$	Circle	Centre $(0, a)$, radius a; passes through origin
$r = a(1 + \cos\theta)$	Cardioid	Cusp at origin, max $r = 2a$ at $\theta = 0$
$r = a\cos n\theta$	Rose (n petals if odd, $2n$ if even)	Each petal length a
$r = a\theta$	Archimedean spiral	Coils outward from origin
$r^2 = a^2\cos 2\theta$	Lemniscate	Figure-of-eight; exists only for $\lvert\theta\rvert \leq \pi/4$







2.2 Sketching Technique

1. Make a table of θ values (e.g., $0, \pi/6, \pi/4, \pi/3, \pi/2, \pi, \ldots$) and compute r
2. Note symmetry: if $r(\theta) = r(-\theta)$, symmetric about initial line; if $r(\theta) = r(\pi - \theta)$, symmetric about $\theta = \pi/2$
3. Note where $r = 0$ (curve passes through origin)
4. Note where r is maximum
5. Plot and connect smoothly

Exam tip: For rose curves $r = a\cos n\theta$: the curve has n petals if n is odd, and $2n$ petals if n is even. Each petal has maximum length a.

3. Area Enclosed by a Polar Curve

The area enclosed by $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$ is:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, d\theta$$

Derivation: A small sector with angle $d\theta$ and radius r has area $\approx \frac{1}{2}r^2 \, d\theta$. Summing gives the integral.

Example: Find the area enclosed by the cardioid $r = a(1 + \cos\theta)$.

By symmetry about $\theta = 0$:

$$A = 2 \cdot \frac{1}{2}\int_0^{\pi} a^2(1+\cos\theta)^2 \, d\theta = a^2 \int_0^{\pi} (1 + 2\cos\theta + \cos^2\theta) \, d\theta$$

Using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$:

$$= a^2 \int_0^{\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{\cos 2\theta}{2}\right) d\theta$$

$$= a^2 \left[\frac{3\theta}{2} + 2\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi} = a^2\left(\frac{3\pi}{2}\right) = \frac{3\pi a^2}{2}$$

Example: Find the area of one petal of $r = a\cos 2\theta$.

One petal in $-\pi/4 \le \theta \le \pi/4$ (where $\cos 2\theta \ge 0$):

$$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} a^2\cos^2 2\theta \, d\theta = a^2 \int_0^{\pi/4} \cos^2 2\theta \, d\theta$$

$$= a^2 \int_0^{\pi/4} \frac{1+\cos 4\theta}{2} \, d\theta = a^2 \left[\frac{\theta}{2} + \frac{\sin 4\theta}{8}\right]_0^{\pi/4} = a^2 \cdot \frac{\pi}{8} = \frac{\pi a^2}{8}$$

4. Area Between Two Polar Curves

The area between $r = f(\theta)$ (outer) and $r = g(\theta)$ (inner) for $\alpha \le \theta \le \beta$:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} \left[f(\theta)^2 - g(\theta)^2\right]\, d\theta = \frac{1}{2}\int_{\alpha}^{\beta} (r_1^2 - r_2^2)\, d\theta$$

Example: Find the area inside $r = 2\cos\theta$ but outside $r = 1$.

Curves intersect where $2\cos\theta = 1$, i.e. $\theta = \pm\pi/3$.
By symmetry:

$$A = 2 \cdot \frac{1}{2}\int_{0}^{\pi/3} \left[(2\cos\theta)^2 - 1^2\right]\, d\theta = \int_{0}^{\pi/3} (4\cos^2\theta - 1)\, d\theta$$

$$= \int_{0}^{\pi/3} (2 + 2\cos 2\theta - 1)\, d\theta = \int_{0}^{\pi/3} (1 + 2\cos 2\theta)\, d\theta$$

$$= \left[\theta + \sin 2\theta\right]_0^{\pi/3} = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

Exam tip: Always find intersection points first by solving $r_1 = r_2$ (and check $r = 0$ cases). Sketch the curves to identify which is outer/inner and the correct range of θ.

5. Converting Polar Equations to Cartesian

Example: Show that $r = 2a\cos\theta$ is a circle in Cartesian form.

Multiply both sides by r: $r^2 = 2ar\cos\theta$.
Substitute: $x^2 + y^2 = 2ax$, i.e. $(x - a)^2 + y^2 = a^2$. Circle, centre $(a, 0)$, radius a. ✓

Example: Convert $r = \dfrac{3}{2 - \cos\theta}$ to Cartesian.

$r(2 - \cos\theta) = 3 \Rightarrow 2r - r\cos\theta = 3 \Rightarrow 2\sqrt{x^2 + y^2} - x = 3$
$2\sqrt{x^2 + y^2} = x + 3 \Rightarrow 4(x^2 + y^2) = (x + 3)^2$

6. Tangents to Polar Curves

Convert to parametric with θ as parameter: $x = r\cos\theta$, $y = r\sin\theta$.

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

where:

$$\frac{dy}{d\theta} = \frac{dr}{d\theta}\sin\theta + r\cos\theta, \qquad \frac{dx}{d\theta} = \frac{dr}{d\theta}\cos\theta - r\sin\theta$$

Tangents parallel to initial line ($y = 0$ direction, i.e. horizontal): set $\dfrac{dy}{d\theta} = 0$ and $\dfrac{dx}{d\theta} \neq 0$.

Tangents perpendicular to initial line (vertical): set $\dfrac{dx}{d\theta} = 0$ and $\dfrac{dy}{d\theta} \neq 0$.

Example: For the cardioid $r = a(1 + \cos\theta)$, find $\dfrac{dy}{dx}$ and the angles at which tangents are parallel to the initial line.

$\dfrac{dr}{d\theta} = -a\sin\theta.$
$\dfrac{dy}{d\theta} = -a\sin\theta \cdot \sin\theta + a(1 + \cos\theta)\cos\theta = a(\cos\theta + \cos 2\theta)$
Set $\dfrac{dy}{d\theta} = 0$: $\cos\theta + \cos 2\theta = 0$, i.e. $\cos\theta + 2\cos^2\theta - 1 = 0$.
Let $c = \cos\theta$: $2c^2 + c - 1 = 0 \Rightarrow (2c - 1)(c + 1) = 0$.
$c = 1/2$ ($\theta = \pi/3, 5\pi/3$) or $c = -1$ ($\theta = \pi$, but check $dx/d\theta$ at this point).

7. Exam-Style Practice Questions

1. **[Conversion]** Convert the polar equation $r = 4\sin\theta$ to Cartesian form and describe the curve.
2. **[Area]** Find the total area enclosed by the curve $r = 2 + 3\sin\theta$ between $\theta = 0$ and $\theta = 2\pi$.
3. **[Area — rose]** Find the total area enclosed by all the petals of $r = 4\cos 3\theta$.
4. **[Area between curves]** Find the area of the region that is inside $r = 3\cos\theta$ and outside $r = 1 + \cos\theta$.
5. **[Tangent]** For the curve $r = 1 - \cos\theta$, find the Cartesian equation of the tangent at $\theta = \pi/2$.
6. **[Sketch]** Sketch the curve $r = 2 + \cos\theta$, labelling key features. State the range of r.

General exam tips for CP2.5:

- Area formula: $\frac{1}{2}\int r^2\,d\theta$. The $\frac{1}{2}$ is frequently forgotten — costs all subsequent marks.
- Use symmetry where possible — halve the interval and double — saves integration work.
- For tangent questions: differentiate $y = r\sin\theta$ and $x = r\cos\theta$ with respect to θ using product rule.
- Lemniscate: exists only for $|\cos 2\theta| \leq 1$, i.e. only certain θ ranges give real r.

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